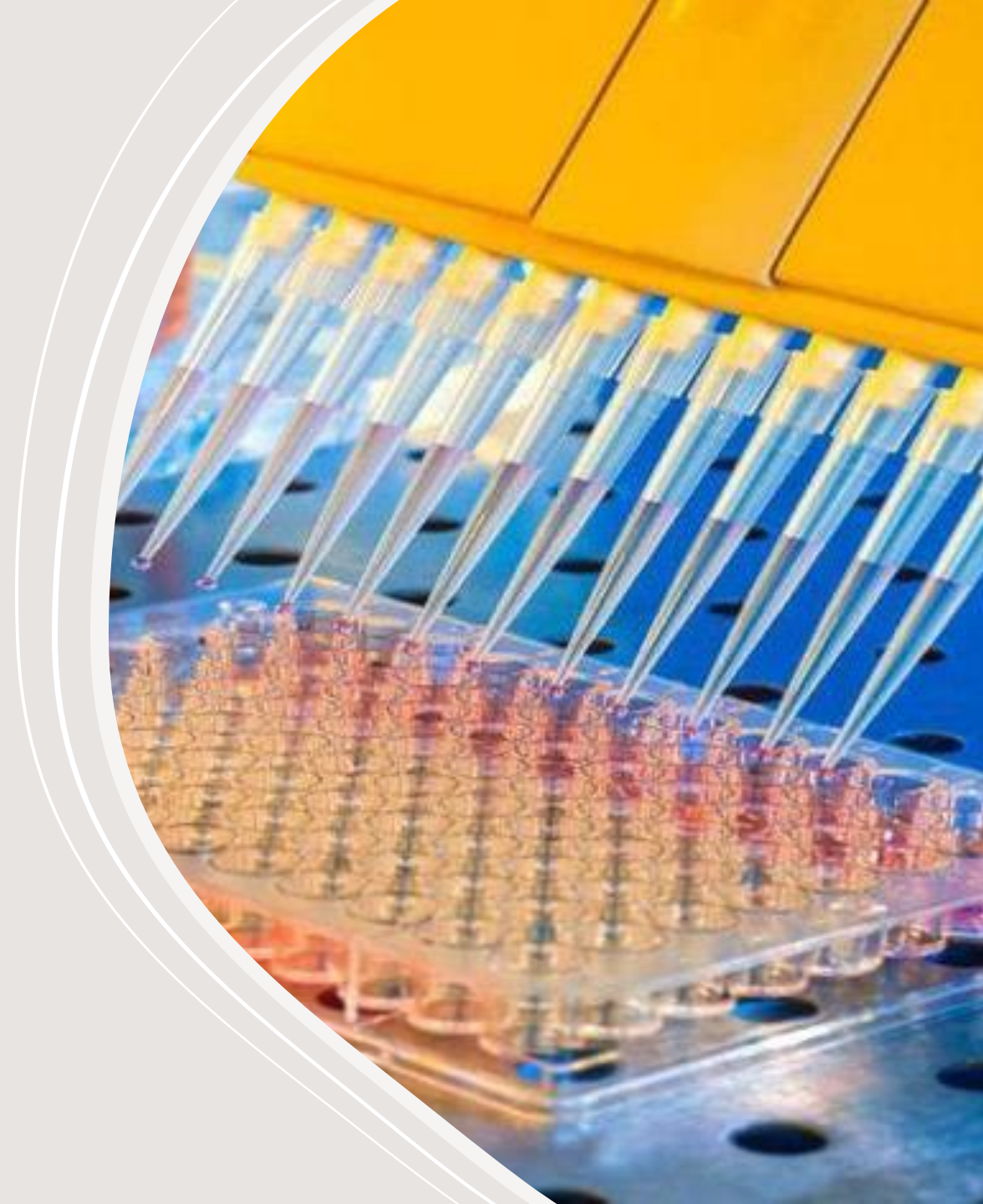


ELISA



Screening Test

- The main purpose of screening test • select group of B cells secreting desired Abs out of Millions of Irrelevant cells produce by the injected mice or generate by fusion process.
- Measurement of the desired specific Antibodies.
- Factors consider in screening test :
 - Sensitivity
 - Specificity
 - Speed

Some techniques are used to Abs screening:

- **Double immunodiffusion** : non sensitive technique, not suitable for Ab screening.
- **Radioimmunoassay**: sensitive but dangerous.
- **Enzyme linked immunosorbant Assay (ELISA)** : the best technique for Ab screening, it is high sensitivity, high specificity, more accuracy and less dangerous.

Monoclonal production

- 1.Immunization of mice
- 2.Screening for Ab production by ELISA
- 3.Isolation of B cells from spleen of injected mice
- 4.Preparation of myeloma cells in tissue culture
- 5.Fusion of myeloma cells with the isolated B cells

Screening

6. Propagation of hybridoma cells in culture.

Screening

7. Expansion of hybridoma cells.

Screening

8. Cloning of hybridoma cells.

Screening

9. Large scale production of Mabs.

Screening

10. Characterization of Mabs: isotype determination
11. Characterization of Mabs: Titer determination

- 
- After each step in monoclonal production the screening is essential.

Sensitivity :

Sensitivity is the lowest value of analyte in assay buffer where the assay can statistically differentiate from background (control).

- Ability of device to differentiate between zero and very low concentration of analyte.
- Some devices can detect milligrams, if used to measure something in micrograms the

device read= zero.

- 
- Some devices detect in Micrograms range, if used to measure something in Nano gram range,

device read = zero

- ELISA is highly sensitive technique, detect Nano grams by using high affinity Abs to improve sensitive.
- Sensitivity require high specificity and low background.

Specificity :

- ELISA must be specific for the target and specificity confirm by capture Ab.
- The robust ELISA use Antibodies don't cross – react non target molecules, closely related metabolic product and Homogeneous interfering molecules • using Monoclonal Antibodies to detect the target.
- Target could be :

1. Ag

2. Ab

- 
- Human Albumin, OVA Albumin, BSA → contain common epitopes.
 - Human Albumin ELISA kit : using Monoclonal Antibody against unique epitope only present in Human Albumin.
 - Human Albumin ELISA don't react with Ova Albumin or bovine serum Albumin (BSA).
 - ELISA use validate (Highly specific Ab) against target molecule.

Precision and Reproducibility :

- Order to have confidence in the results obtained from a single assay plate, or to be able to compare results obtained from different plates, it is critical to test the assay's Precision and Reproducibility.
- Intra-Assay Precision is a measurement of reproducibility between wells within an assay. This allows the researcher to run multiple replicates of the same sample on one plate and obtain similar results. Typically, intra-assay precision is tested by running three samples (triplicates).

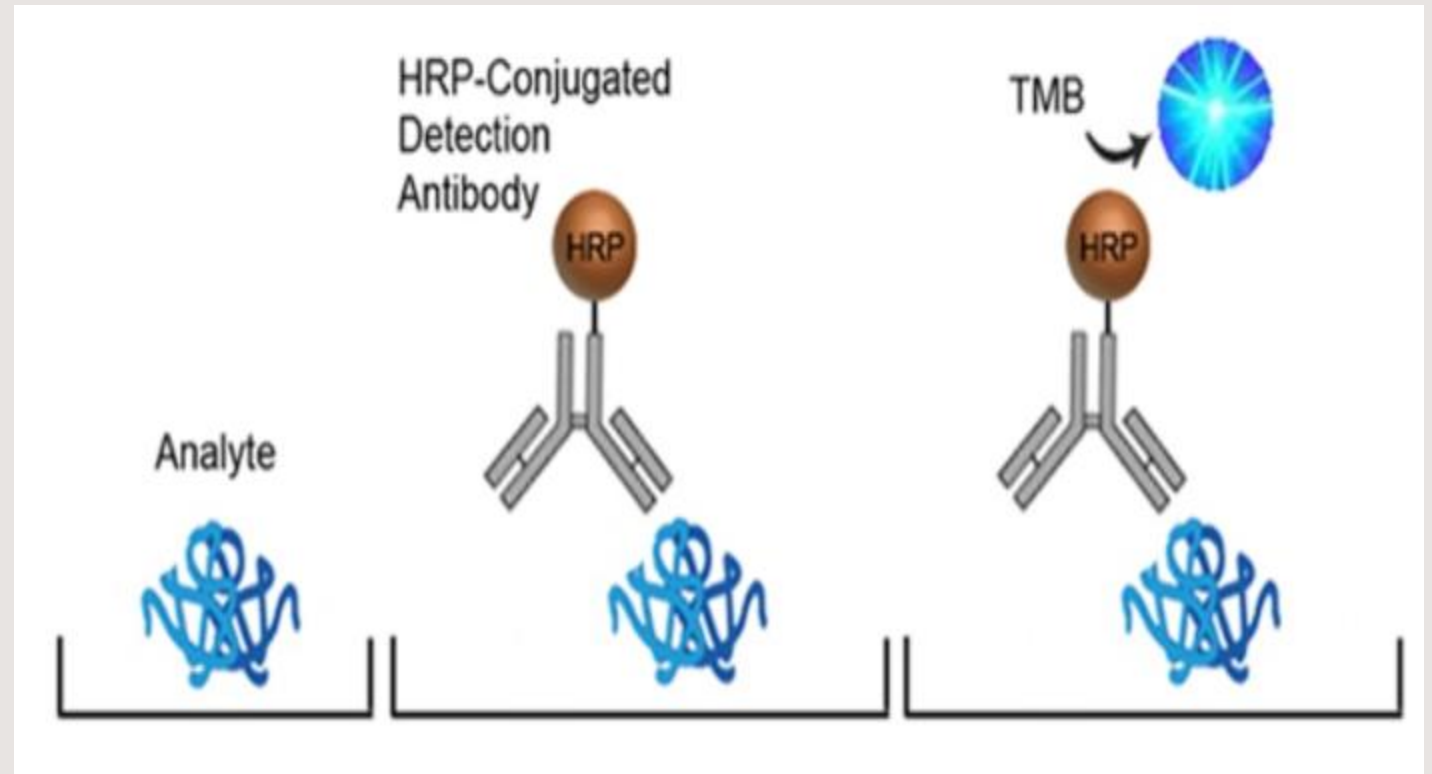


- Inter-Assay Precision is a measurement of reproducibility between the wells of different assay plates over time. This gives the researcher the ability to compare results obtained from different plates run at different times. Typically, inter-assay precision is tested by running three samples on three different plates at different times.

- In both cases, Precision is measured as a coefficient of variation (CV) from the mean value. Ideal CV values are $<10\%$ for Intra-assay Precision and $<15\%$ for Inter-assay Precision.

Types of ELISA

- To ensure the Immunization was successful • Antibodies against the specific Antigen must Exist.
- Now the target is→ looking for Antibodies against the specific Antigen in mouse serum.
- **Direct ELISA**
 - In direct ELISA, wells coat with Anti-body to detect the Antigen.
 - Not suitable for Mab screening.



- **Capture Ab ELISA / sandwich ELISA.**

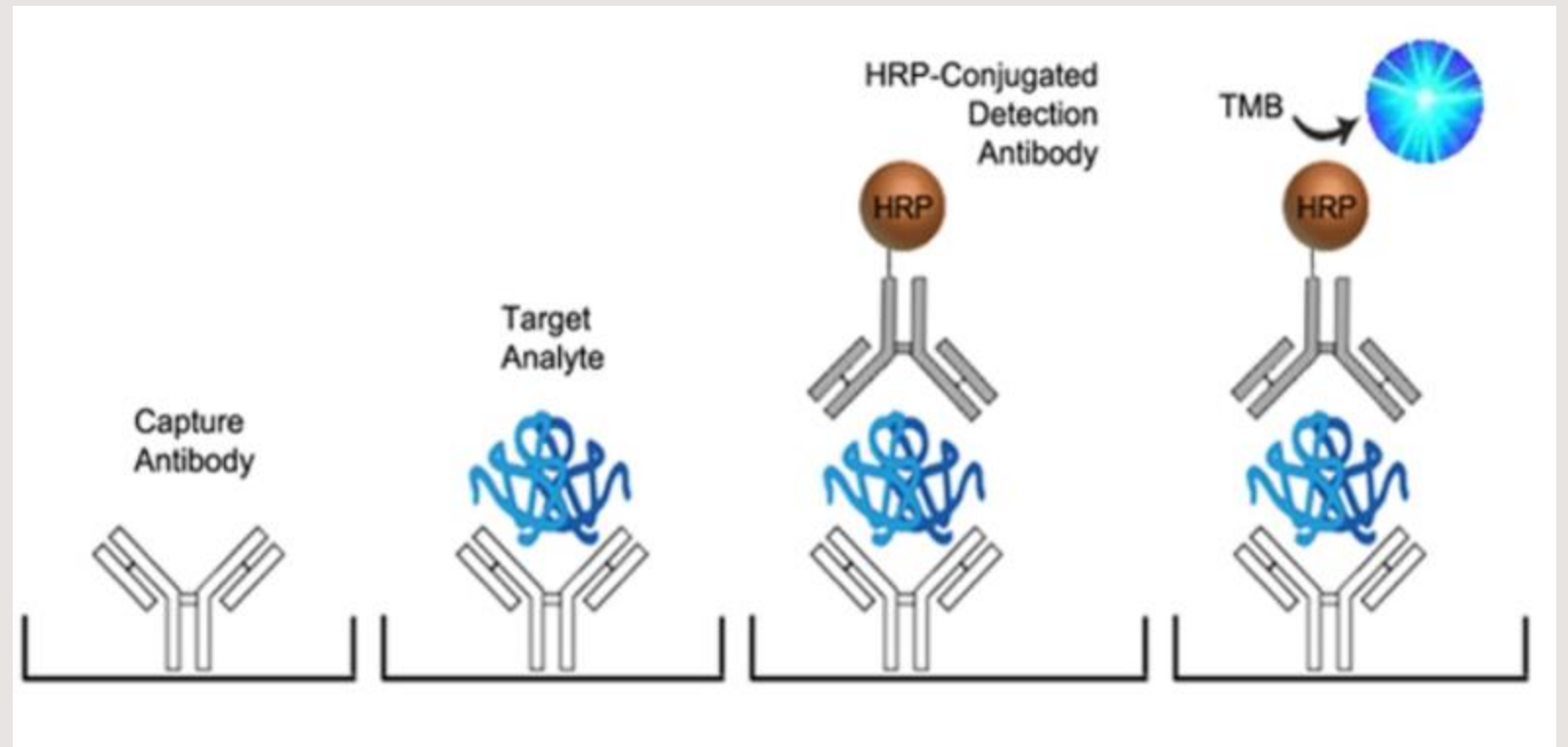
- Wells coat with Antibodies to detect the target Antigen.

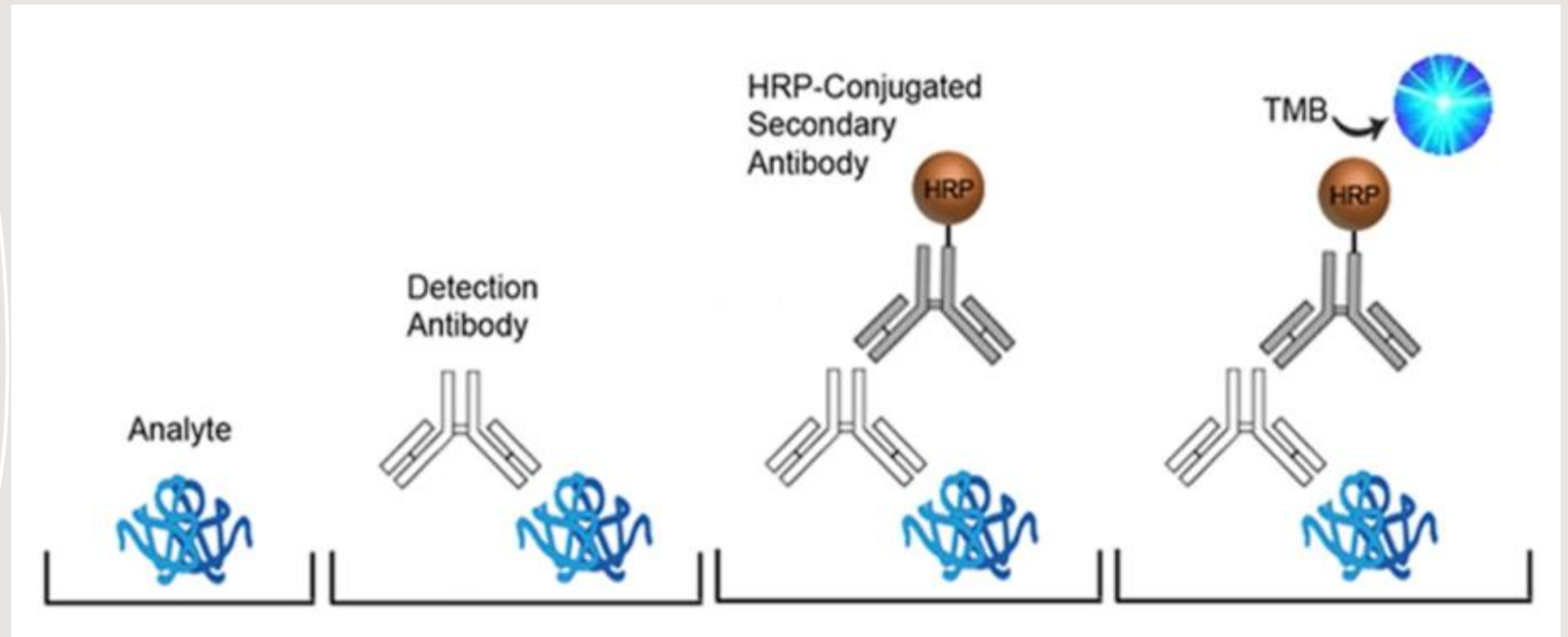
- Not suitable for Mab screening.

- **In directly ELISA**

- Wells coat with Antigens to detect the target Antibodies.

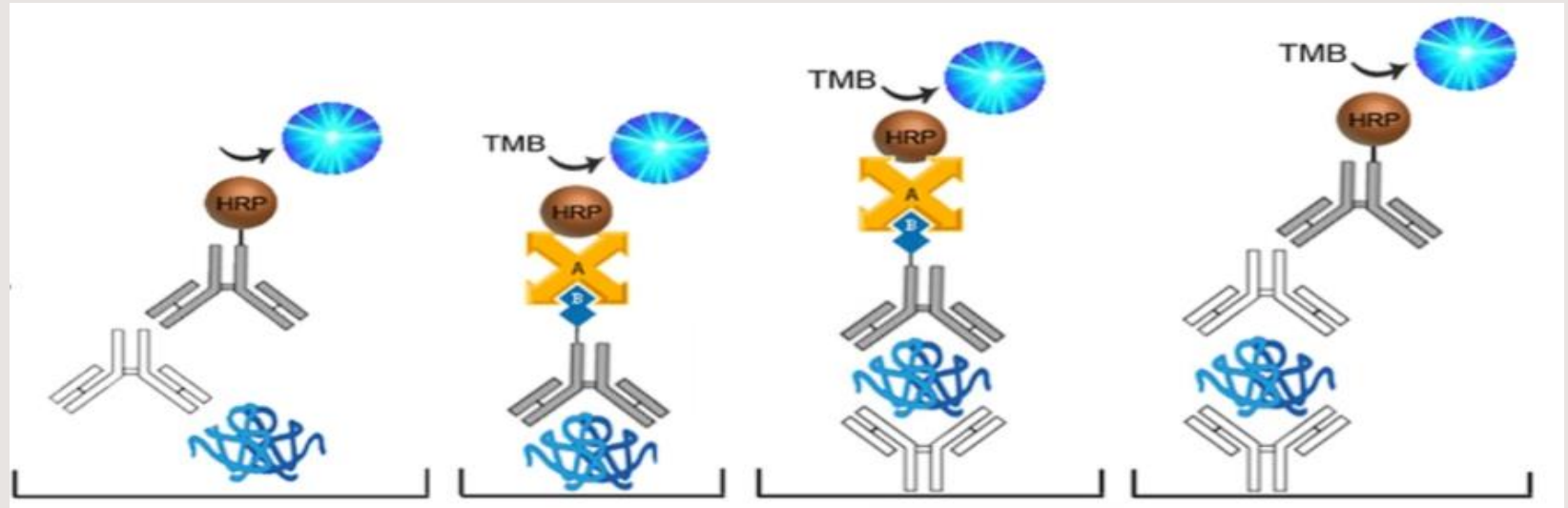
- Suitable for Mab screening.

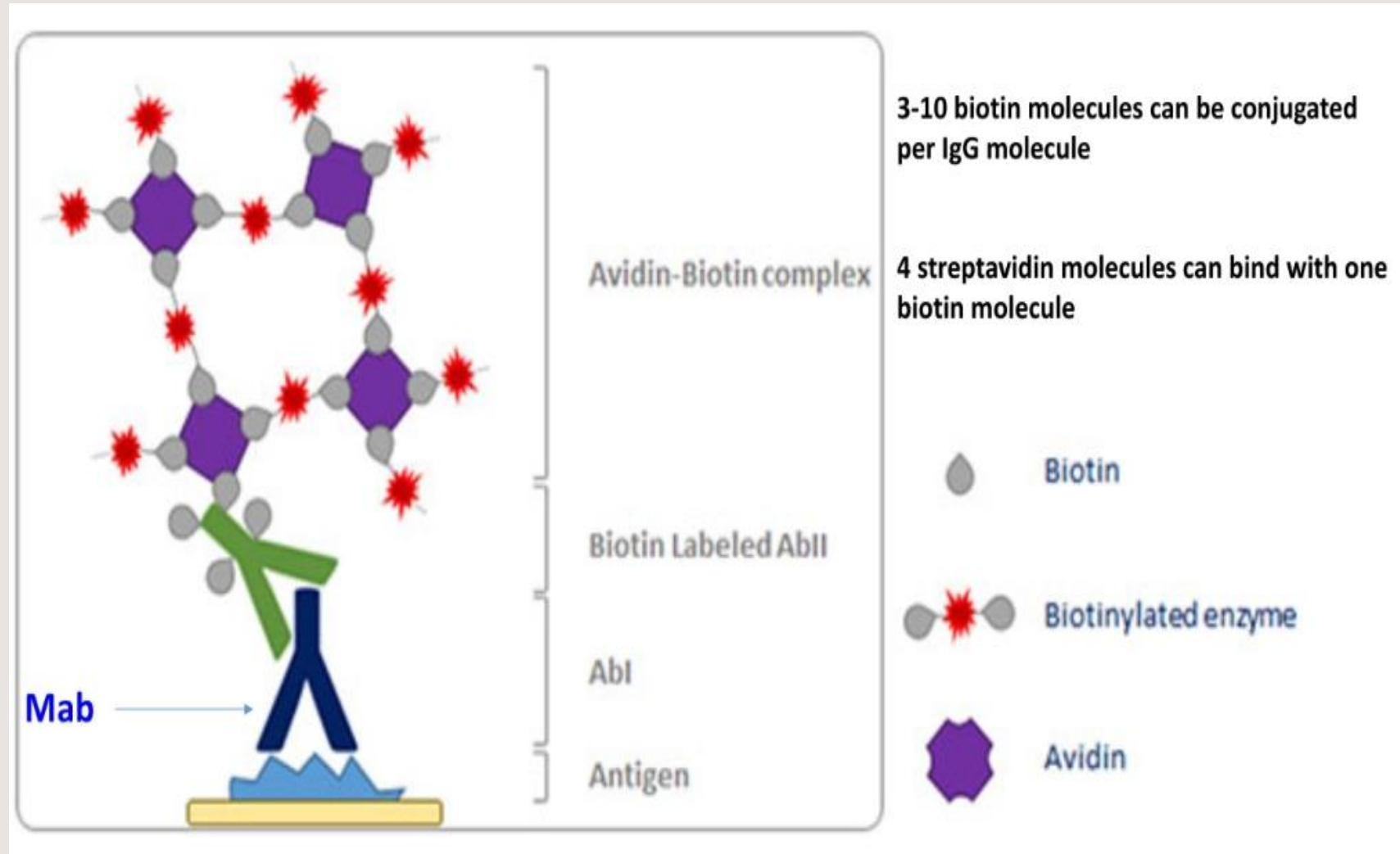




Avidin-biotin ELISA :

- In direct ELISA, Amplification can be achieved by Avidin-biotin system using biotinylated secondary Abs followed by streptavidin-HRP improves the sensitivity.
- The biotin-streptavidin interaction offers a 4:1 increase in signal, since 4 streptavidin molecules can bind with the biotin.
- Suitable for Mab screening.



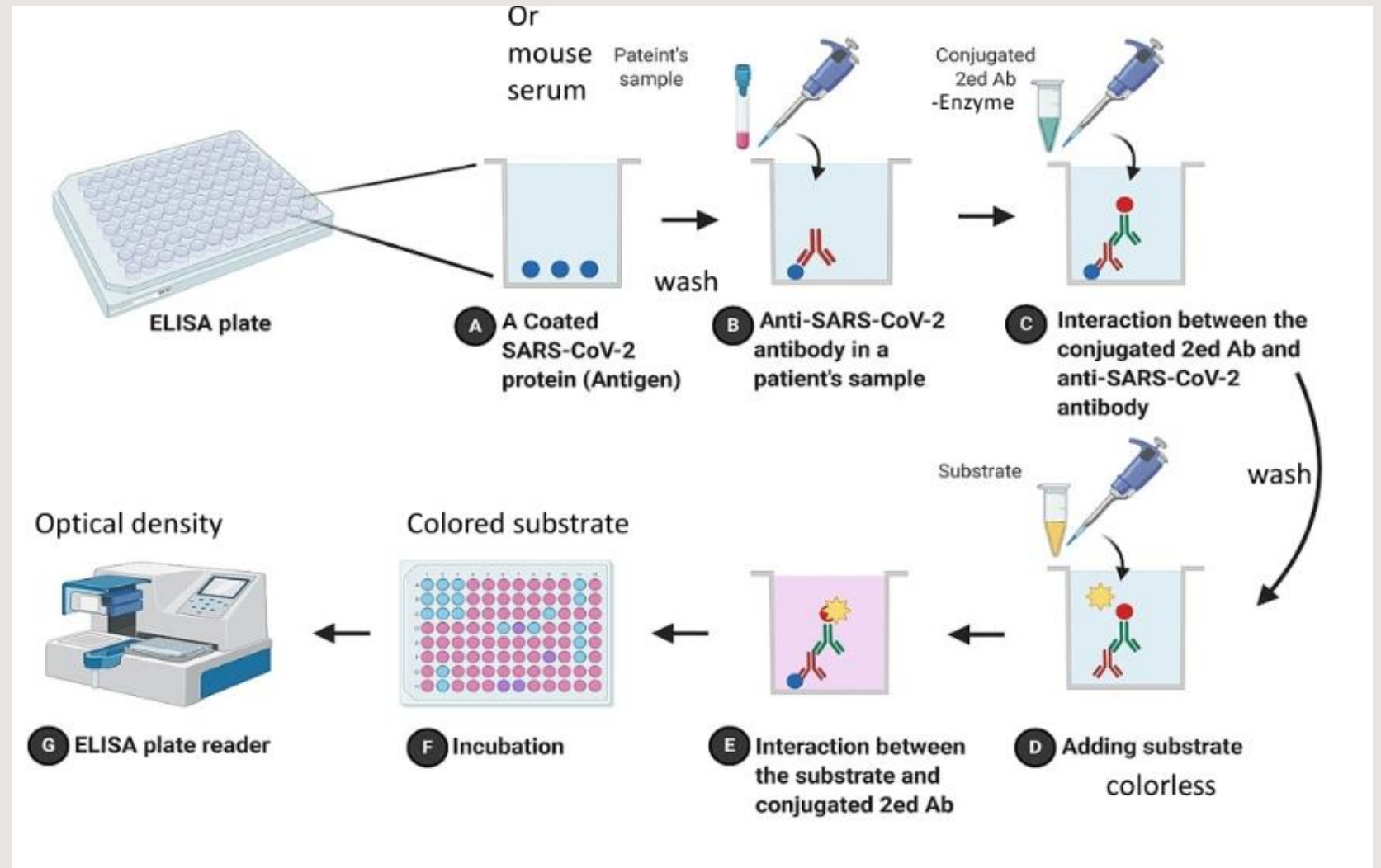


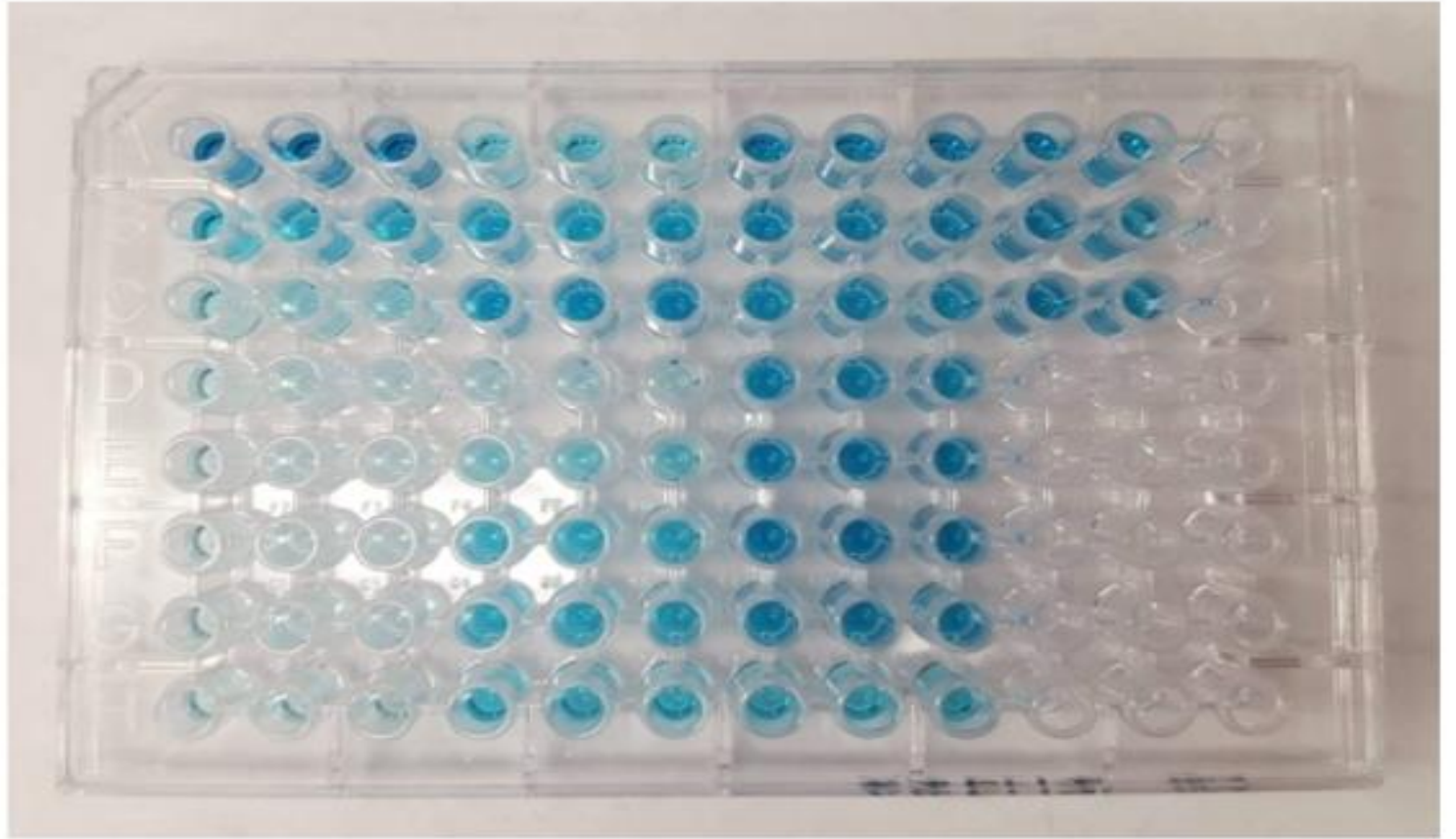
ELISA steps

1. Ag coating
2. Washing
3. Test sample addition or controls (positive, negative and others)
4. Washing
5. Enzyme-labelled anti-mouse Ig conjugate
6. Washing
7. Substrate
8. Stopping solution
9. Absorption using ELISA reader

Microtiter plates

- Solid support usually from polystyrene, Non specific bind protein Antigens (bind many Antigen not specific for one Antigen) with high affinity via Adhesion to the surface.
- Strip wells: provide greatest flexibility, allowing use only wells need for Experiment rather than all whole plate.
- Fixed wells: using Whole Microtiter plate all 96 wells at the same time.





Microtiter plate 96 wells

Bottom Shapes Of 96-Well Plates :

- The 96-well format microplates are available in four distinct bottom shapes. .

1.U-bottom: round-shaped well bottom, used in agglutination assay.

2.V-bottom: conically-shaped well bottom, used in precipitation reactions.

3.F-Bottom: flat well-bottom, used in ELISA and tissue culture

Ag coating:

- Ag must be pure to reduce false positive result

1. Ag must be soluble, concentration of Ag add 10-100 micrograms / ml

- Too little Ag add : false negative result
- Too much Ag add : formation of multilayers of Ag which lead to unstable bonding.

2.Using Buffer without detergents

- carbonate buffer (the best)
- Tris buffer (2)
- Phosphate buffer (3)

3.Incubation overnight at 4c°, make sure the plate sealed to minimize the effect of evaporation, can kept for weeks.

- If the coating Antigen cells (not proteins)
- Attached to plastic surfaces via bridging molecules which has a high affinity for the plastic e.g Glutaraldehyde or PHA.

Blocking solution

- After coated the Plate with target Ag, Maybe there is empty Sites, these sites are available for binding for any Ag Remember the plate surface non specific for binding target Ag, Maybe closely Related Ag bind in these sites and cause false positive result.
- To overcome this problem→ use block solution, block solution contain Irrelevant proteins (don't have common epitopes) to occupy all the empty sites, to minimize the false positive result.



- If the test samples are culture supernatants accumulated from dying cells, it is possible that the debris will contribute to nonspecific binding (false positive). This effect can be minimized by changing the culture medium a few days before screening.

- Common proteins use as blocking solution

- BSA

- Ova Albumin

Washing

- Actually washing is essential after each step in ELISA.
- use Tween 20 (0.05%) for Washing.
- Usually sample wash 3 time.
- In efficient washing cause non specific variability.

Test serum

- Ab levels in mouse serum or culture supernatant rarely exceed 10ug/ml so they are normally tested without dilution
- Most of the wells screened will be negative for specific Ab
- Specific Ab levels can be expected to be double or even treble the background.
- Background levels are likely to increase as the incubation period increases.

Enzyme Ab Conjugate (secondary Ab) :

- Ab conjugation, also known as Ab labeling, is the process of linking an Ab to a specific tag or label. Most immunoassays require Abs either directly conjugated to a specific label or detected by a conjugated secondary Ab to provide a measurable signal.
- Secondary Ab label by link to enzyme (HRP).

- 
- After addition of Ab Conjugate to wells, washing, then add Substrate for the enzyme (OPD or TMP), when the substrate add the color is changed.

OPD→ orange

TMP→yellow

HRP inactive at 15c°

- Not advisable when using plant tissues due to endogenous HR.
- Alkaline phosphatase AP is a good alternative with PNPP as substrate (avoid phosphate buffer).

Controls

TO have reliable results

1-To test for nonspecific Ab

- a. Coat alternative Ag and add the same tested Ab
- b. Add irrelevant Ab to the same coated Ag

2- To quantify specific binding

→Add known positive Ab

3-To test for nonspecific conjugate label

→Add buffer.

